Symbol Communication with modulo ADC

Krunal Vaghela

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1 Introduction

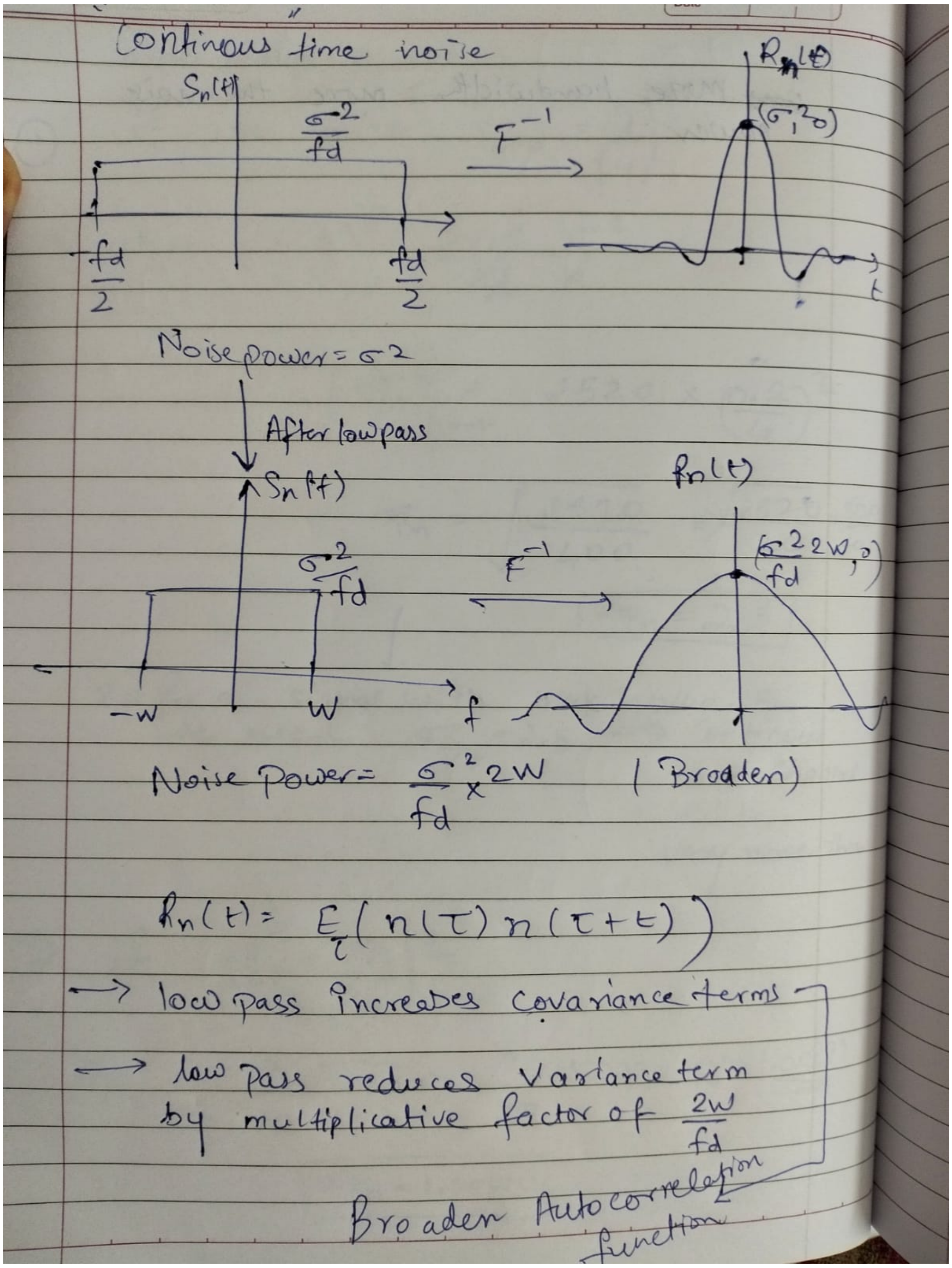

Figure 1: Effect of LPF

Noise variance after LPF

The variance of a stationary noise process equals the area under its PSD:

$$\sigma_n^2 = \int_{-\infty}^{\infty} S_n(f)\, df \tag{1}$$

Given the original PSD

$$S_n(f) = \begin{cases} \dfrac{\sigma^2}{F_d}, & |f| \le \dfrac{F_d}{2} \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

After a low-pass filter with bandwidth W ($W < F_d/2$):

$$\sigma_{\text{out}}^2 = \int_{-W}^{W} \frac{\sigma^2}{F_d}\, df \tag{3}$$

$$\sigma_{\text{out}}^2 = \frac{\sigma^2}{F_d}(2W) \tag{4}$$

$$\boxed{\sigma_{\text{out}}^2 = \sigma^2 \frac{2W}{F_d}} \tag{5}$$

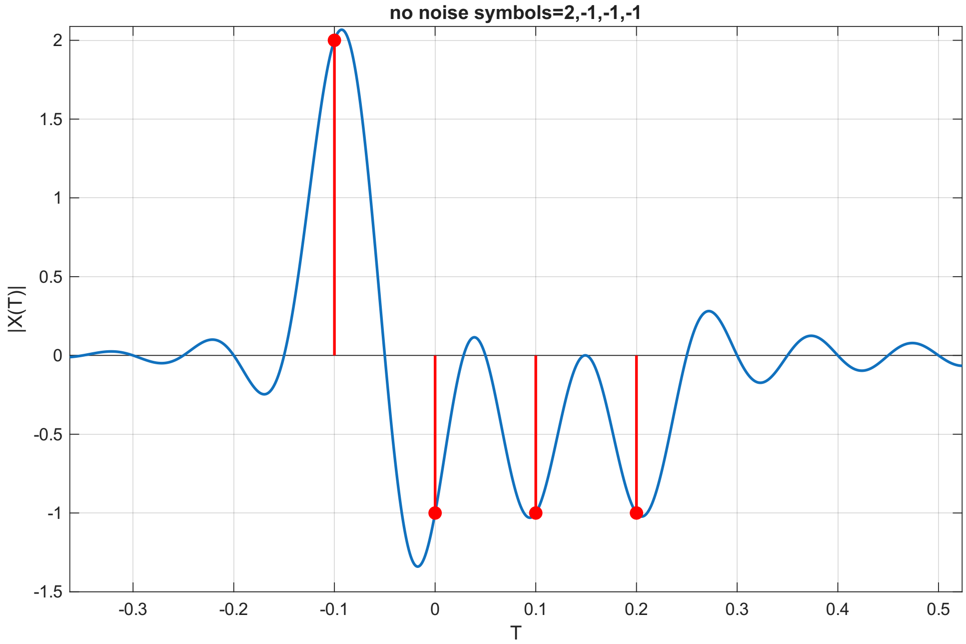

Figure 2: original signal

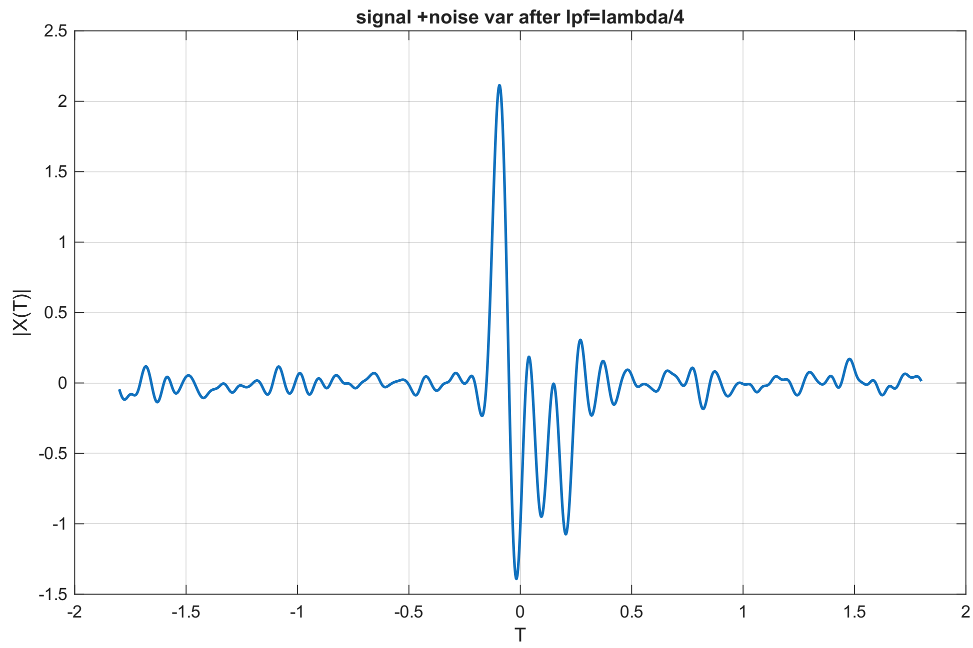

Figure 3: signal + lpf noise

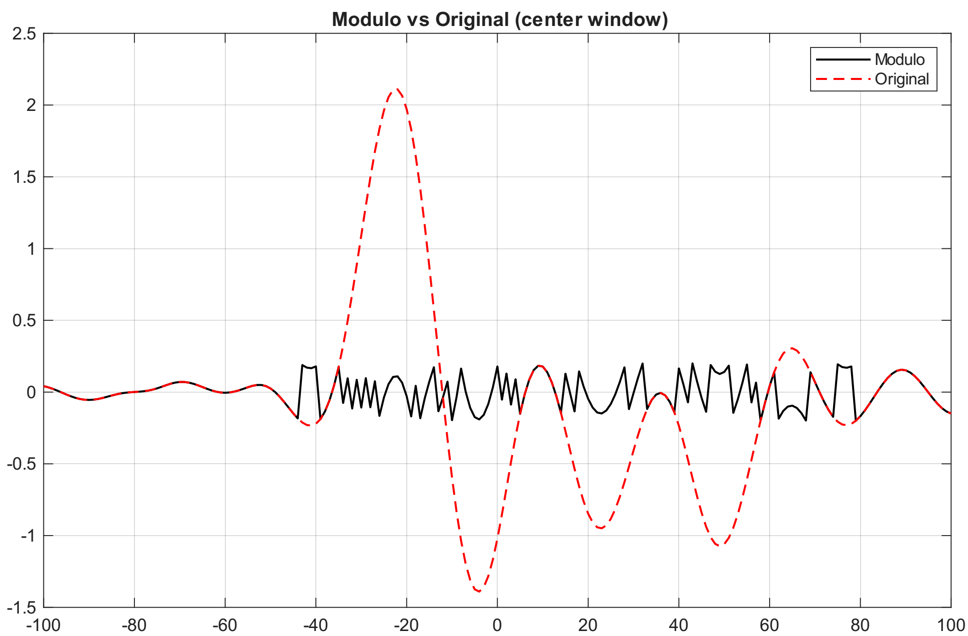

Figure 4: folded

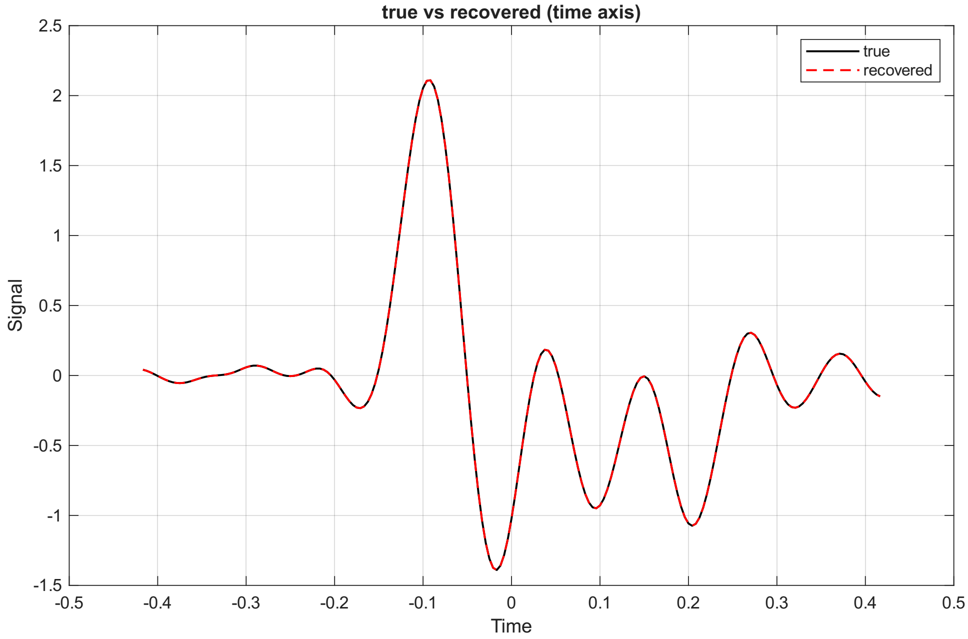

Figure 5: reconstructed

2 MSE , SER and probability of recovery plots at different noise variance after lowpass ,lambda=0.2, S=[1,-1,2,-2]

3 $\sigma_{LPF}^2 = (Lambda/10)^2, maxsignalvalue = c + 3 * \lambda/10 = 2 + 0.06 = 2.06$

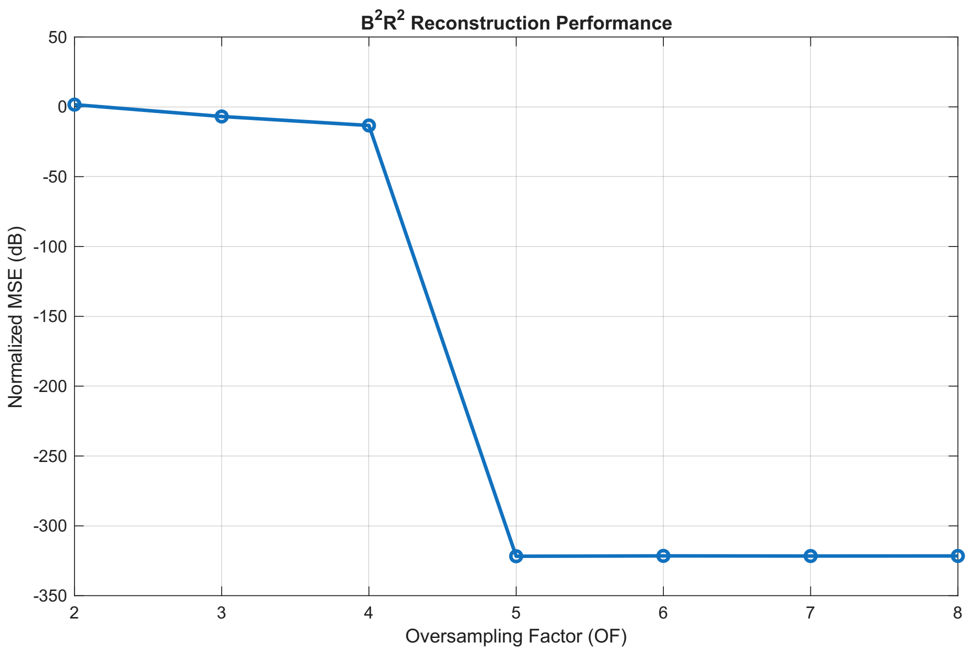

Figure 6: MSE

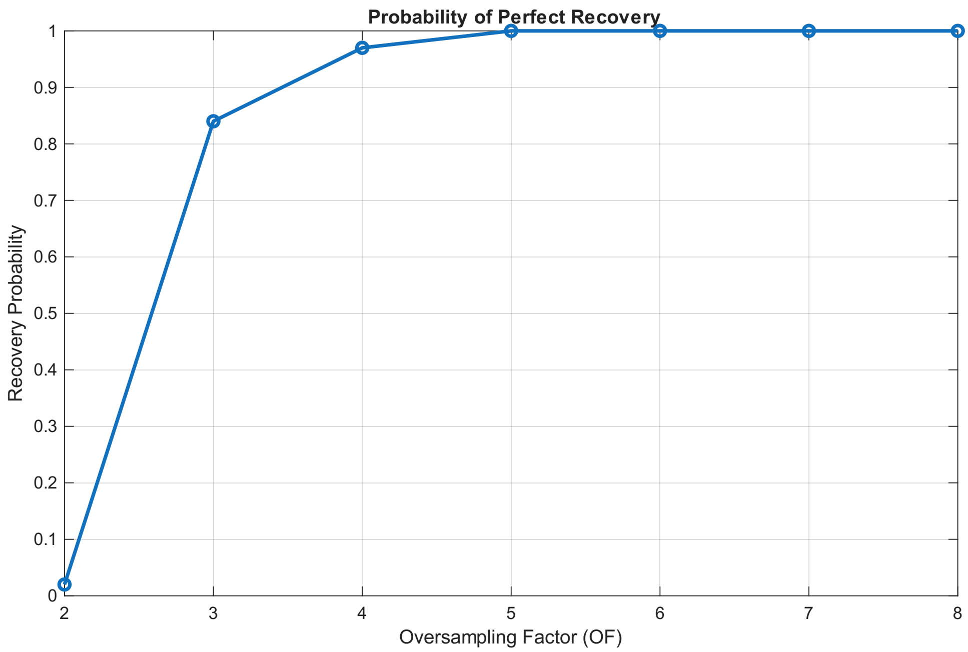

Figure 7: PE

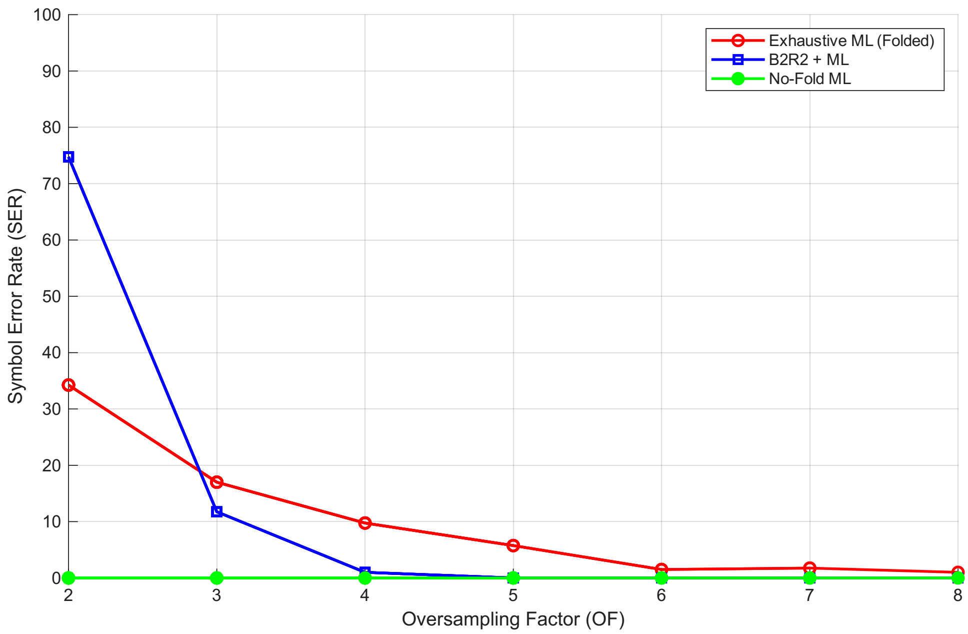

Figure 8: SER

4 $\sigma^2 = (lambda/8)^2, maxvalueofsignal = 2 + 0.075 = 2.075$

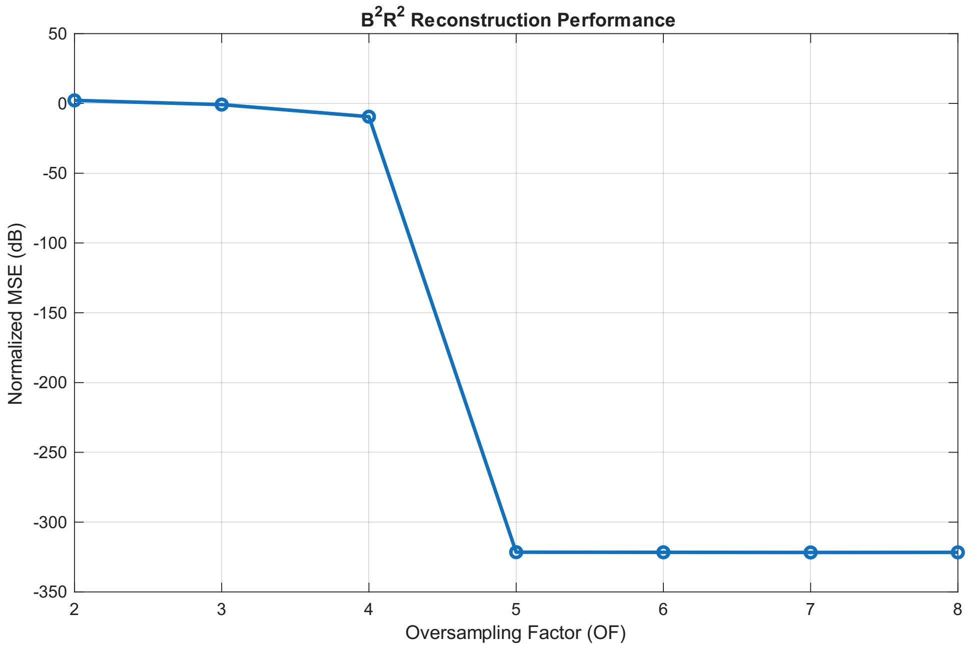

Figure 9: mse

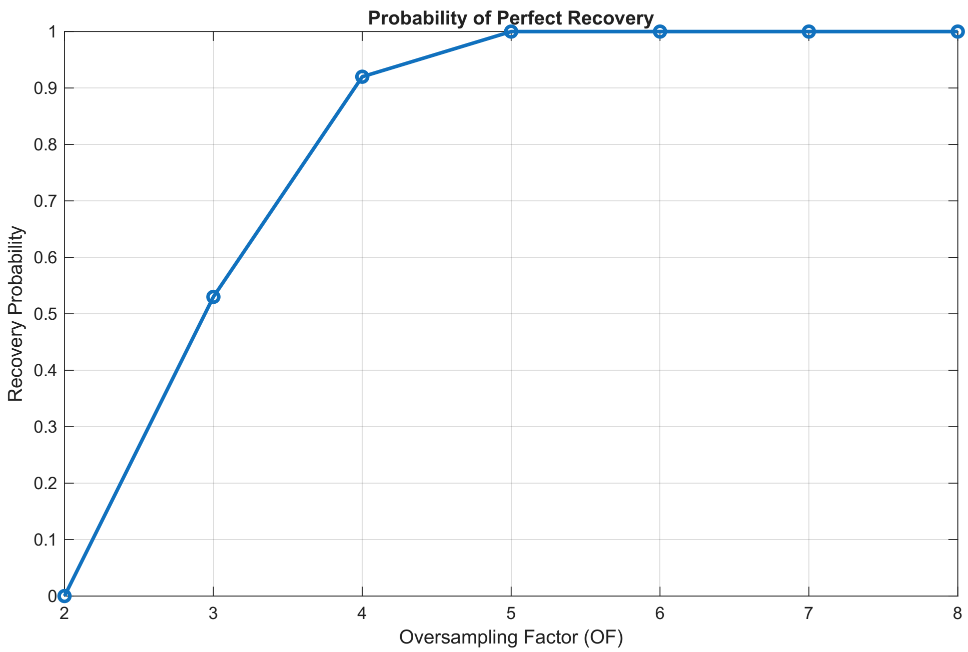

Figure 10: probablity error

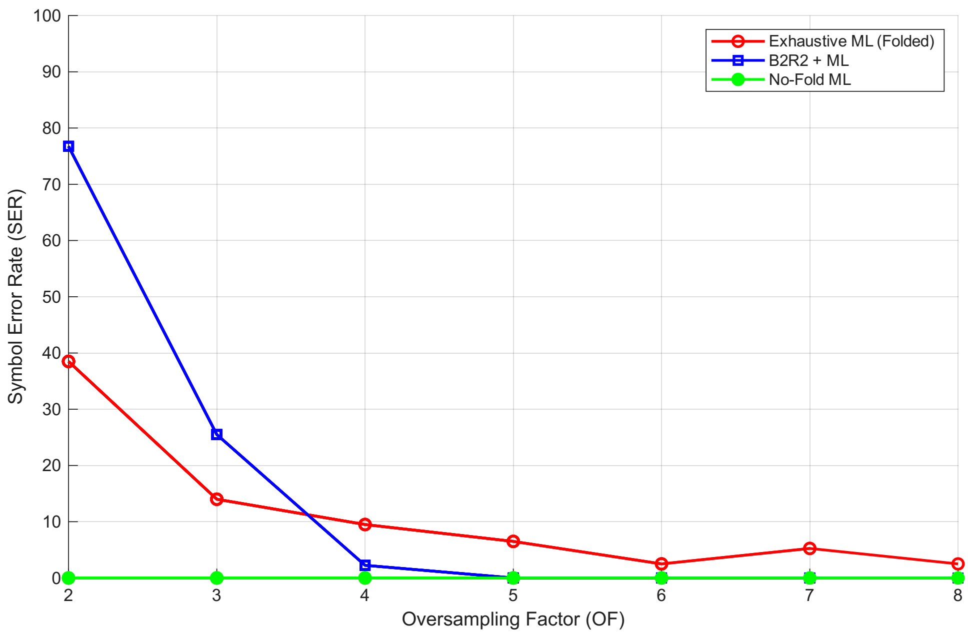

Figure 11: SER

5 sigma=lambda/6, max value of signal = 2.1

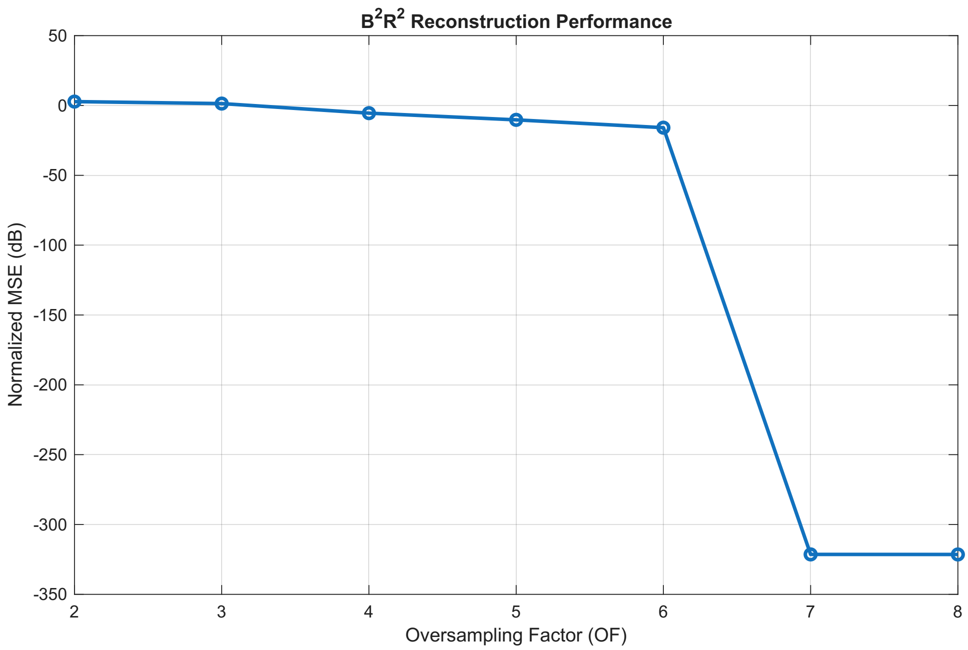

Figure 12: mse

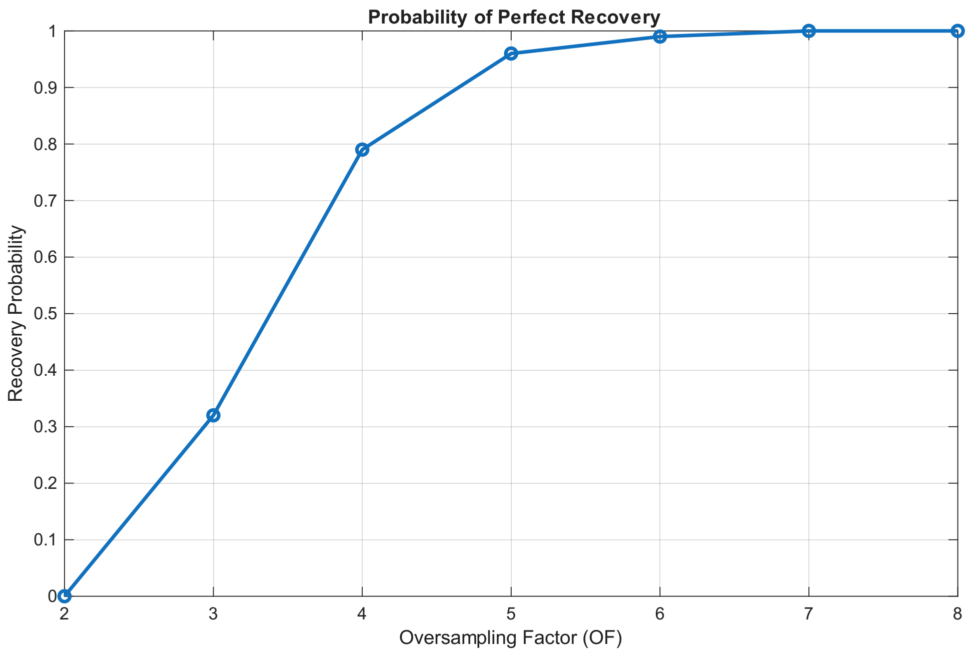

Figure 13: pe

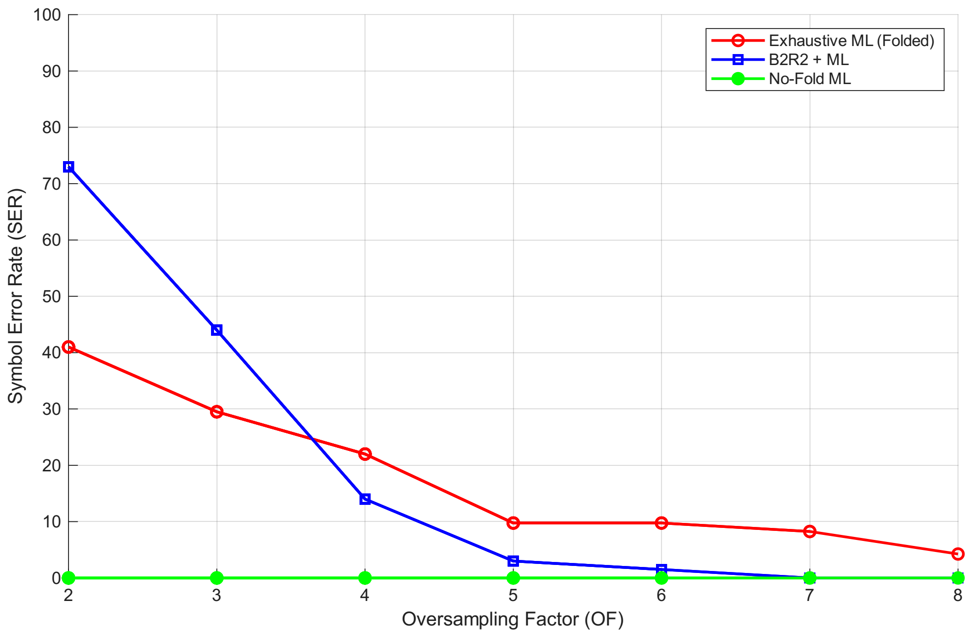

Figure 14: SER

6 sigma=lambda/5, max value of signal=2.12

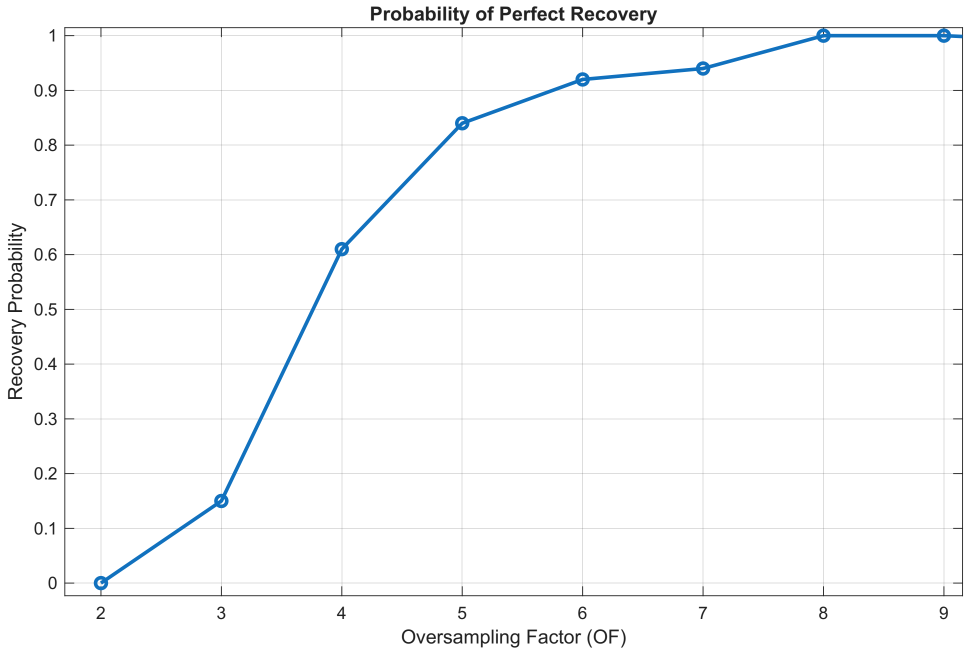

Figure 15: pe

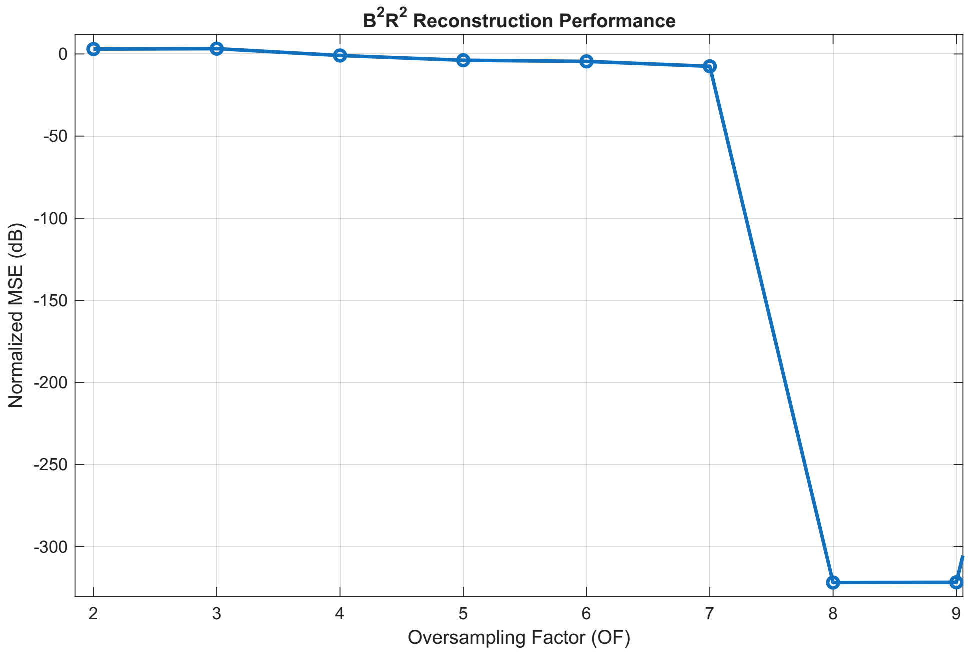

Figure 16: mse

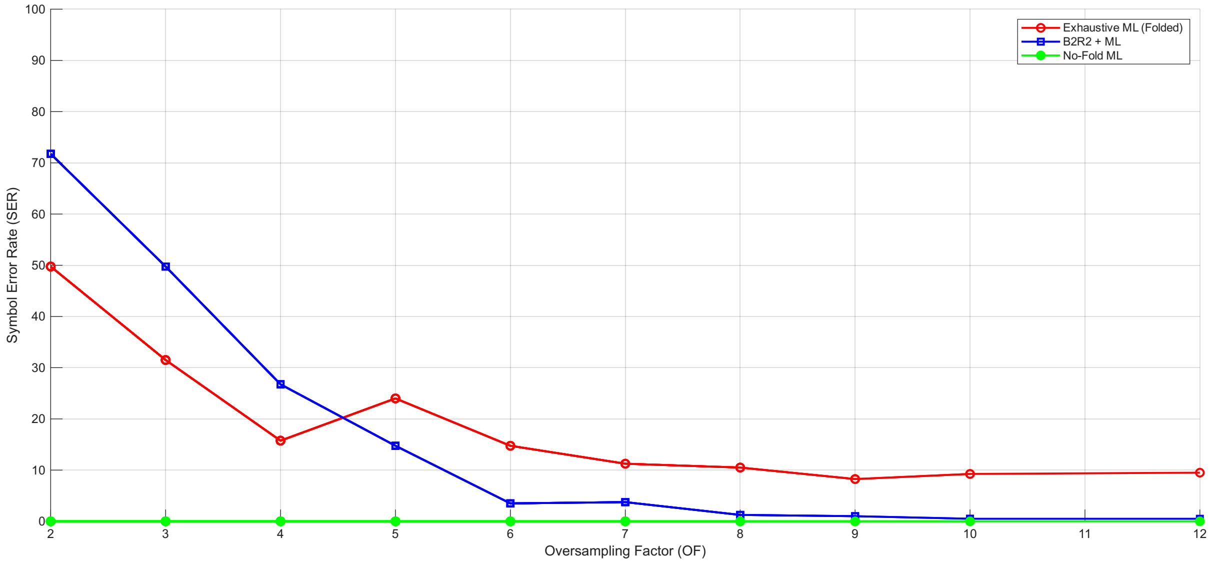

Figure 17: ser

7 $\sigma^2_{LPF} = (Lambda/4)^2, max value of signal = 2.15$

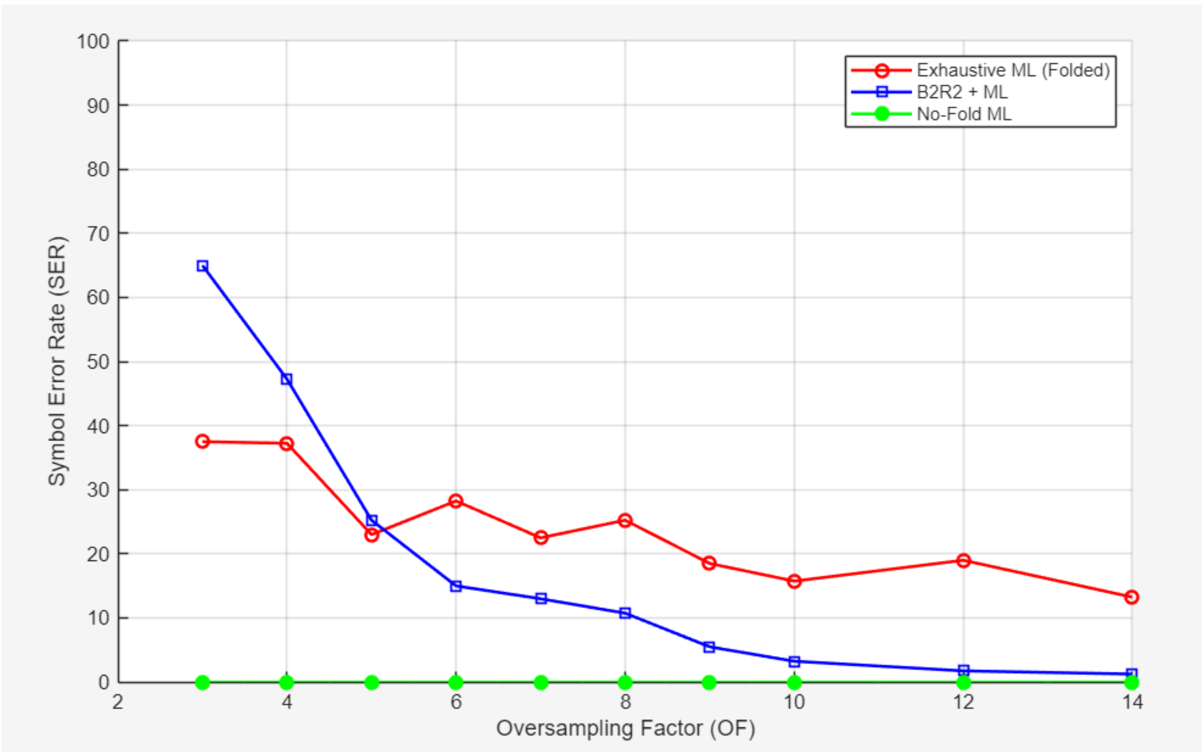

Figure 18: SER

Previous Mistakes

B2R2 showed unusual oversampling trends due to improper signal generation.

$$T_d = 10^{-4} \tag{6}$$

Continuous time grid:

$$t_d = [-1 : T_d : 1] \tag{7}$$

Sampling period:

$$T_s = \frac{2\pi}{\omega_s} \tag{8}$$

Previously the sampling indices were computed as

$$\text{step} = \text{round}\left(\frac{T_s}{T_d}\right) \tag{9}$$

$$\text{for } i = 1 : \text{end} \tag{10}$$

$$t_s = t_d[i] \tag{11}$$

$$i = i + \text{step} \tag{12}$$

The rounding operation created inconsistencies in the sampling locations.

Improved Approach: Downsampling Method

Instead of computing sample indices using rounding, a dense continuous-time grid is first generated and then downsampled.

$$T_d = \frac{T_s}{1000} \tag{13}$$

Note that T_d depends on the oversampling factor:

$$T_d = \frac{1}{(2\,OF\,f_m)\,1000} \tag{14}$$

The dense time grid becomes

$$t_d = [-1 : T_d : 1] \tag{15}$$

As the oversampling factor increases, the number of continuous-time samples between two sampled points remains fixed at 1000.

The sampled signal is then obtained using

$$t_s = \text{downsample}(t_d, 1000) \tag{16}$$

Thus, the sampling instants remain fixed while the intermediate dense samples are removed. The so-called continuous-time noise is implemented as discrete-time noise with very high temporal resolution (small T_d).

$$F_d = \frac{1}{T_d} \tag{17}$$

where F_d represents the effective continuous-time sampling frequency of the generated noise.

The noise is then passed through a low-pass filter to limit its bandwidth:

$$\text{noise}_{\text{LPF}} = \text{lowpass}(\text{noise},\ W,\ F_d) \tag{18}$$

where W denotes the signal bandwidth.

Noise Power Analysis

The generated noise has a flat power spectral density

$$S_n(f) = \begin{cases} \dfrac{\sigma^2}{F_d}, & |f| \leq \dfrac{F_d}{2} \\ 0, & \text{otherwise} \end{cases} \tag{19}$$

The total noise variance equals the area under the PSD:

$$\sigma^2 = \frac{\sigma^2}{F_d} \cdot F_d \tag{20}$$

After low-pass filtering with bandwidth W:

$$\sigma^2_{\text{out}} = \int_{-W}^{W} \frac{\sigma^2}{F_d}\, df \tag{21}$$

$$\sigma^2_{\text{out}} = \frac{\sigma^2}{F_d}(2W) \tag{22}$$

$$\sigma^2_{\text{out}} = \sigma^2\,(2WT_d) \tag{23}$$

Noise Power After LPF

$$\sigma^2_{\text{LPF}} = \frac{\sigma^2}{OF} \tag{24}$$

$$\boxed{\text{Noise power after LPF} = \frac{\sigma^2}{OF}} \tag{25}$$

However, increasing the continuous-time resolution with increasing oversampling factor OF is not physically meaningful.
Hence, the dense-time resolution T_d should be kept constant regardless of OF.

Previously,

$$T_d = \frac{T_s}{1000} \tag{26}$$

which varies with OF. Instead we define

$$T_d = \frac{T_s}{\alpha} \tag{27}$$

where α is an integer that changes with OF.
To maintain a fixed continuous-time resolution,

$$\alpha = \frac{1}{2\,OF\,W\,T_d} \tag{28}$$

The dense time grid is generated as

$$t_d = [-1 : T_d : 1] \tag{29}$$

The sampled grid is obtained via downsampling

$$t_s = \text{downsample}(t_d, \alpha) \tag{30}$$

where

$$\alpha \in \mathbb{Z} \tag{31}$$

i.e., α must be an integer. Let the dense-time resolution be

$$T_d = \frac{1}{2WK} \tag{32}$$

Then the downsampling factor becomes

$$\alpha = \frac{1}{2\,OF\,W\,T_d} \tag{33}$$

Substituting T_d:

$$\alpha = \frac{1}{2\,OF\,W\left(\frac{1}{2WK}\right)} = \frac{K}{OF} \tag{34}$$

For α to be an integer, K must be a multiple of OF.

Suppose we want to simulate performance for

$$OF = 2, 3, 4, \ldots, 10$$

Then K must be divisible by all these numbers. Hence we choose

$$K = \mathrm{LCM}(2, 3, 4, 5, 6, 7, 8, 9, 10) \tag{35}$$

$$K = 2520 \tag{36}$$

This ensures that T_d remains constant while the downsampling factor α remains integer for every oversampling factor.

Noise Power After LPF

$$\sigma^2_{\text{LPF}} = \frac{(2W)\sigma^2}{F_d} \tag{37}$$

Since

$$F_d = 2WK \tag{38}$$

we obtain

$$\sigma^2_{\text{LPF}} = \frac{2W\sigma^2}{2WK} = \frac{\sigma^2}{K} \tag{39}$$

Thus the noise power becomes

$$\boxed{\sigma^2_{\text{LPF}} = \frac{\sigma^2}{K}} \tag{40}$$

which is constant and does not depend on the oversampling factor.

8 NEXT tasks

Can we make b2r2 work with variable sampling location since our information of interest is peak value of sinc ,can we sample more near the sinc peaks and get better SER at lower OFs.

9 Including exhaustive search in b2r2 loop

While $\mathrm{N}_\lambda > 0:\quad \widehat{f} = \widehat{f}_\lambda$

for $\mathrm{k} = 1 : 10^3$

$\mathrm{y}^k = z^{k-1} - \gamma_k\, \mathbf{F}_\rho^* \mathbf{F}_p \left(z^{k-1} - \widehat{f} \right)$

$\mathrm{z}^k = \mathcal{P}_{\mathcal{S}_N}\left(y^k \right)$

$\mathrm{a}^k = \arg\min_{a \in \mathcal{A}} \left\| \widehat{f} - \left(H a^{k-1} + z^k \right) \right\|^2$

$\widehat{f} = \text{modulo}\left(H a^k \right)$

end

10 Partial DFT Model and Nullspace-Based Decomposition

Let $z[n]$ be a discrete-time signal of length N. We assume that only a subset of its frequency samples are observed.
$\mathrm{f}_\lambda[nT_s] = f[nT_s] + z[nT_s]$

10.1 DFT Representation

The DFT of $z[n]$ is given by:

$$F(e^{j\omega_k}) = \sum_{n=0}^{N-1} z[n] e^{-j\omega_k n} \tag{41}$$

This can be written in matrix form as:

$$\mathbf{F} = \mathbf{F}_N \mathbf{z} \tag{42}$$

where:

- $\mathbf{F}_N \in \mathbb{C}^{N \times N}$ is the DFT matrix
- $\mathbf{z} \in \mathbb{C}^{N \times 1}$ is the signal

—

10.2 Partial DFT Observation

We observe only a subset of frequency components:

$$\mathbf{F}_\lambda = \mathbf{A}\mathbf{z} \tag{43}$$

where:

- $\mathbf{A} \in \mathbb{C}^{m \times N}$ is a selection matrix (partial DFT)
- $m < N$

—

10.3 Signal Decomposition

We decompose the signal into:

$$\mathbf{z} = \mathbf{z}_T + \mathbf{z}_C \tag{44}$$

where:

- $\mathbf{z}_T$: components in the transform (observed) subspace
- $\mathbf{z}_C$: components in the complementary subspace

Thus:

$$\mathbf{F}_\lambda = \mathbf{A}_T \mathbf{z}_T + \mathbf{A}_C \mathbf{z}_C \tag{45}$$

—

10.4 Nullspace Property

Let the SVD of $\mathbf{A}_C$ be:

$$\mathbf{A}_C = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H \tag{46}$$

Partition $\mathbf{U}$ as:

$$\mathbf{U} = [\mathbf{U}_1 \ \mathbf{U}_2] \tag{47}$$

Then:

$$\mathbf{U}_2^H \mathbf{A}_C = 0 \tag{48}$$

This implies that $\mathbf{U}_2$ spans the nullspace of $\mathbf{A}_C$.
—

10.5 Projection onto Nullspace

Multiplying both sides:

$$\mathbf{U}_2^H \mathbf{F}_\lambda = \mathbf{U}_2^H \mathbf{A}_T \mathbf{z}_T \tag{49}$$

—

10.6 Recovery of Unknown Component

We solve:

$$\mathbf{z}_T = (\mathbf{A}_T^H \mathbf{A}_T)^{-1} \mathbf{A}_T^H \mathbf{F}_\lambda \tag{50}$$